

Q.1. What and Why PCA?

- It is feature reduction techniques used to solve curse dimensionality reduction problem .
- It simplifies large datasets by reducing a large collection of variables into a smaller set that retains most of the information from the original set.

Q.2 What is the curse of dimensionality?

- The curse of dimensionality refers to problems that arise when analyzing data in high-dimensional spaces. As the number of features increases, the complexity of the model also increases, leading to potential overfitting. This means the model performs well on training data but poorly on new, unseen data, because it becomes too dependent on the training data.

Q.3 How PCA works ?

- Steps:

- 1. Define the Data:

Start with your dataset, which has n samples and m features.

- 2. Scale the Features:

Standardize the data (e.g., using min-max scaling) so that each feature has the same scale.

- 3. Compute the Covariance Matrix:

Calculate the covariance matrix to understand how the features vary together.

- 4. Calculate Eigenvalues and Eigenvectors:

Determine the eigenvalues and eigenvectors of the covariance matrix. Eigenvalues show the amount of variance captured by each principal component, and eigenvectors define the

direction of these components

5. Select Principal Components:

Sort the eigenvalues in descending order and select the top k eigenvectors (Principal components) corresponding to the largest eigenvalues.

6. Project the Data:

Transform the original data onto the new axes formed by the selected principal components. This results in a reduced dataset with k dimensions.

Ex:

Student Math Science English

A 85 78 90

B 72 65 80

C 90 85 88

D 60 70 75

E 88 90 92

1. Define the data in matrix form :

[

[85, 78, 90],

[72, 65, 80],

[90, 85, 88],

[60, 70, 75],

[88, 90, 92]

]

2. Standardize the Features: Scale the data so that each feature has the same scale.

3. Compute the Covariance Matrix: Calculate the covariance matrix to understand how the features vary together.

Covariance matrix=

[
[1, 0.66, 0.82],
[0.66, 1, 0.81],
[0.82, 0.81, 1]
]

4. Calculate Eigenvalues and Eigenvectors: Determine the eigenvalues and eigenvectors of the covariance matrix.

Eigenvalues = [2.55, 0.39, 0.06]

Eigenvectors =

[
[0.58, -0.8, 0.15],
[0.57, 0.54, -0.61],
[0.59, 0.27, 0.75]
]

5. Sort the eigenvalues in descending order and select the top k eigenvectors corresponding to the largest eigenvalues.

Selected eigenvectors (principal components):

[
[0.58, -0.8],
[0.57, 0.54],
[0.59, 0.27]
]

6. Project the Data: Transform the original data onto the new axes formed by the selected principal components.

Transformed data =

[
[0.59, -0.42],
[-1.29, 0.38],
[0.91, -0.13],
[-1.72, 0.45],

[1.51, -0.29]

]

- By applying PCA, we reduced the dimensionality of the students' test scores from 3 to 2 while retaining most of the information.